Question #1: First Question (15 marks): Fill-in the table with the correct answer:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
C	B	B	B	B	D	A	D	C	A	A	B

1. If the size of memory is 512 bytes, what will be the starting and ending offset addresses of it?

A. 00H to FFH B. 000H to 1FFH C. 0000H to FFFFH D. 000H to FFFH

2. What is the type of addressing mode used in instruction MOV AL, [BP+6]

A. Indirect B. Based relative C. Direct Indexed D. Direct

3. the word length of Intel 8086 microprocessor is:

A. 24 B. 16 C. 20 D. 32

16

4. 5. If the offset address of X in data segment is equal to 118H, and DS=2000H, then the physical address of X =?

20000+118
20118

A. 21118 H B. 20118H C. 21117 H D. 31117H

5. Which is part of execution unit in Intel 8086 microprocessor?

A. Segments registers B. ALU C. physical address generating unit D. IP register

6. Show the binary value in register AL after execution the following instruction sequence,

```
MOV AL,11D   -> Ah = 11    0B
MOV BL,15D   Bh = 15 -> 0F
SUB AL,BL    -> Ah = 0B - 0F ->
```

A. AL=11111111 B. AL=11111011 C. AL=11111110 D. AL=11111100

7. If the SP register contains the hexadecimal value F11FH, what will be its value after executing the following program segment?

```
PUSH AX
PUSH BX
ADD AX,BX
POP AX
POP AX
PUSH CX
```


A. F11FH B. F11AH C. F11CH D. F11DH

8. Find the value of AX after execution the following assembler codes

```
MOV CX,5       -> CX = 0005
MOV AX,0       AX = 0000
L: ADD AX,CX   -> ADD = 00 + 05 = 0005    AX = 0005 + 0004 = 0009 , AX = 0012
DEC CX         -> CX = 0005 - 1 -> 0004   , CX = 0004 - 0001 = 0003   CX = 0002
CMP CX,2       -> 0004 > 2                                            stop
JA L
HLT
```

A. 9 B. 15 C. 8 D. 12

9. Which is not part of execution unit in Intel 8086 microprocessor?

A. Segments registers B. ALU C. physical address generating unit D. IP register

10. The content of register BL after execution the instruction MOV BL,-11

A. BL=F6H B. BL=F7H C. BL=0BHH D. BL=06H

11. Find the content of AL and zero flag (ZF) and carry flag (SF) after execution of the following code:

```
MOV AL, 0FDH
ADD AL, 3        -> AL = 00   CF=1
ADC AL, 0FEH     -> AL = 00+1 -> 0   ZF=0   SF=1
```

A. AL=FFH,ZF=0;SF=1   B. ~~AL=00H,ZF=0;SF=1~~   C. AL=FFH,ZF=0;SF=0   C. AL=02H,ZF=0;SF=1

12. The offset of a particular segment varies from ----------to----------

A.000h to FFFh   B. 0000h to FFFFh   C. 00h to FFh   D. 00000h to FFFFFh

Question #2 (8 points)

Trace out the following program segments and find registers contents:

```
ARR DW 3333H, 4444H, 1111H, 2222H, 0FFFH

MOV BX,OFFSET ARR      BX = CX
MOV SI, 4        -> SI=4
MOV DI,2            DI=2
MOV DL, [BX ][SI]   DL = X+4   DL=11h
MOV AL, [BX]     -> AL = X  ->  AL=33
XCHG AL, DL      ->

AL= 11h ; DL= 33h
```

```
MOV AL, 22H            AL=22h
SUB AL,22              AL= 22-22=0   ZF=zero
JZ YES      -> ZF=1
SUB AL,5
JMP EE
YES: ADD AL,5          AL= 0+5 = 05
EE:HLT

AL= 05h
```

```
ARRAY DB 8, 3, 2, 4,6,8,5,7,0
MOV AL,0                 AL=0
MOV CX,4                 CX=4
MOV BX, OFFSET ARRAY     BX=[A]
L:
ADD AL, [BX]      -> AL = 0+08
ADD BX,2          -> BX = X+2
LOOP L    CX=3, CX=2, CX=1, CX=0    0015
  MOV AH,[BX]

AL= 15  AH= 00h
```

08
03
02
04
06
08
05
07
00

```
X DW 104FH, 201AH, C030H
MOV BX, OFFSET X       BX = CX
MOV AX, [BX]      -> AX = 104F
INC BX            -> X+1
MOV CX,[BX]       -> CX = X+1
PUSH AX           -> 201A
PUSH CX           -> C030
POP AX            -> C030
POP CX            -> 201A

AX= ~~104F~~  CX= ~~1A20~~
AX= C030h  201Ah
```

4F
10
1A
20
30
C0

Question #3(10 points)

a) Suppose we have a microprocessor with 32bits address bus; what is the maximum memory capacity will be?

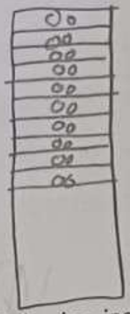
$= 2^{32} = 2^{30} \cdot 2^{2} = 4GB$

b) For the following instruction MOV [BX], 50H; what is the addressing mode used?

INdirect addresing

c) Write an assembler codes to store zeroes into X memory locations (3 points)

X DB 10 DUP (?)



d) Write the equivalent code of the following instructions (2 points)

LOOP L

= ADD CX

JFZ = 1

e) Write an assembler code to implement the following if statement: (3 points)

If X< 8 s=s+15 else s=s+7; assuming x and y are signed bytes.

RoG 100

HLT